# Synthetic Image Generation

https://github.com/aiyshwary/Synthetic_waste_image_generation

Python | Computer Vision | OpenCV | Data Augmentation

## OVERVIEW

An automated pipeline for generating high-quality synthetic training images for object detection, addressing class-imbalanced datasets by compositing annotated object crops onto varied backgrounds with randomized augmentations.

## IMPACT

Built an end-to-end synthetic image generation pipeline that composites real object crops onto diverse backgrounds with randomized augmentations, producing annotated training data that reduces dependency on costly manual data collection.

## TECHNICAL WALKTHROUGH

This project builds an automated pipeline to generate high-quality synthetic images for object detection. The goal is to address limited or imbalanced real- world data by synthesizing new training images using real object crops and annotations.

**Why synthetic data?**

Manual annotation is expensive and time-consuming. Synthetic image generation allows us to increase dataset size, improve class balance, and expose models to diverse object placements without collecting new data.

**End-to-end pipeline explanation**

**Data ingestion from S3**

The pipeline starts by downloading both annotated and unannotated sorted images from an S3 bucket. Duplicate downloads are avoided to ensure efficiency.

**i)** Unannotated images → for background usage

**ii)** Annotated images → for object extraction

**Annotation handling**

For annotated data, CVAT annotations in Datumaro JSON format are downloaded and combined into a single unified annotation file.

**Two paths**

**i)** Convert existing annotations to YOLO format

**ii)** Or auto-annotate using a pre-trained object detection model when annotations are missing

This ensures consistent labeling across the dataset.

**Dataset consolidation**

All images and YOLO labels are flattened into unified folders to standardize downstream processing. This step simplifies:

i) Crop generation

ii) Mask creation

iii) Synthetic placement

**Object crop & mask generation**

From the labeled dataset, object-level crops and corresponding binary masks are generated.

**Outputs**

i) Object crops

ii) Pixel-level masks

iii) Crop-level labels

These form the building blocks for synthetic images.

**Synthetic image generation**

Using the extracted object crops and masks, new images are synthesized by placing objects on background images.

**Two strategies**

i) Spread-out placement → realistic non-overlapping scenes

ii) Clustered placement (optional) → congested scenes with object repetition

This increases:

i) Spatial diversity

ii) Object density variation

iii) Robustness of detection models

**Phase-based synthetic generation**

An optional phase-based generator creates images in multiple stages, gradually increasing object density and class balancing.
This helps:

i) Control dataset distribution

ii) Target specific class counts

iii) Improve long-tail performance

**Visualization & validation**

Finally, synthetic images are visualized with bounding boxes overlaid from YOLO labels to verify annotation quality.
This acts as a sanity check before training.

**Tech stack & reproducibility**

The entire pipeline is Dockerized, ensuring reproducibility across environments.
Dependencies include OpenCV, NumPy, Pandas, and AWS SDK (boto3) for seamless S3 integration.

**One-line summary**

I built a fully automated, Dockerized synthetic data generation pipeline that converts CVAT annotations into YOLO format, extracts object crops, and synthesizes diverse training images to improve object detection performance.

## KEY DETAILS

1. Extracts foreground object crops from existing annotated datasets using bounding box and mask information.

2. Composites crops onto varied background images with randomized scale, rotation, and position transformations.

3. Automatically generates YOLO/Pascal VOC-compatible annotation files alongside each synthetic image.

4. Targets underrepresented classes to balance dataset distributions for improved model generalization.

5. Validated improvements in downstream detection model performance on augmented versus original-only training sets.

6. Uses Ultralytics auto_annotate with a custom YOLO detection model and SAM2 for automatic polygon-level instance segmentation labels.

7. Implements three synthesis strategies: CropScatterSynth (up to 30 objects/image), ClusteredSyntheticGen (40,000 instances/class target), and SyntheticPhaseGen (three-phase YAML-configurable generation) with multithreaded generation.

8. Dockerized on nvidia/cuda:12.2.0 with full 9-step bash pipeline: S3 download, annotation conversion, auto-annotation, crop extraction, synthetic generation, and visualization.